COSC3106: Theory of Computing

CH#3

Lec#7: Regular Expressions

School of Computer Science and Technology

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This lecture

- This lecture, will …
  - Explain regular expression.
  - Describe languages and presenting them in regular expressions.

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What is a regular expressions (RE):

- A Regular Expression is a tool to describes a language of a set of strings over an alphabet.

Ex: This is an example of a regular expression that describes a language over $\sum = \{0, 1\}$

$$\underbrace{(0 \cup 1)}_{\text{Write 0 or 1}}\underbrace{0}_{\text{Write 0}}\underbrace{1}_{\text{Write a finite \# of 1's.}}{}^{*}$$

- " * " Means repeat the string a finite number of times.
- So this language is the set of all bit strings of the following form:

- **Ex#1:** $A = \{w\colon\ w$ **contains exactly two** $0's\}$

- **Ex#2:** $A = \{w\colon\ w$ **contains at least two** $0's\}$

- Ex#3: $A = \{w\text{: } 1011 \text{ is substring of } w\}$

- Ex#4: $A = \{w\text{: length of } w \text{ is even}\}$

- Ex#5: $A = \{w$: length of w is odd $\}$

- Ex#6: $A = \{w$: first bit of w = its last bit $\}$

- The formal definition of regular expressions:
  - Let Σ be a non-empty alphabet.

    1. ε is a regular expression.
    2. $\emptyset$ is a regular expression.
    3. For each $a \in \Sigma$, a is a regular expression.

    Basic cases

    4. If R_1 and R_2 are regular expressions, then $R_1 \cup R_2$ is a regular expression.
    5. If R_1 and R_2 are regular expressions, then $R_1 R_2$ is a regular expression.
    6. If R is a regular expression, then R^* is a regular expression.

- Ex: $\sum = \{0, 1\}$
  - Claim: $(0 \cup 1)^* 101 (0 \cup 1)^*$ is a regular expression.

    How we can argue that this is a regular expression?

From (3): 0 is RE, and 1 is RE

From (4): $(0 \cup 1)$ is RE

From (6): $(0 \cup 1)^*$ is RE

From (5): 10

$$\left.\begin{array}{l} 101 \\ (0 \cup 1)^* 101 \\ (0 \cup 1)^* 101 (0 \cup 1)^* \end{array}\right\} \text{Are REs}$$

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- The regular expressions represent regular languages:
  - Let Σ be a non-empty alphabet.
    1. ε is a regular expression $\Leftrightarrow$ Language: $\{\varepsilon\}$
    2. $\emptyset$ is a regular expression $\Leftrightarrow$ Language: $\{\emptyset\}$
    3. For each $a \in \Sigma$, a is a regular expression $\Leftrightarrow$ Language: $\{a\}$
    4. If R_1 and R_2 are regular expressions, then $R_1 \cup R_2$ is a RE $\Leftrightarrow$ Language: $R_1 \cup R_2$
    5. If R_1 and R_2 are regular expressions, then R_1R_2 is a RE $\Leftrightarrow$ Language: R_1R_2
    6. If R is a regular expression, then R^* is a RE $\Leftrightarrow$ Language: R^*

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- The regular expressions can also be presented in different equivalent representations:
  - Ex: Let $(\mathbf{0} \cup \varepsilon)\mathbf{1}^*$